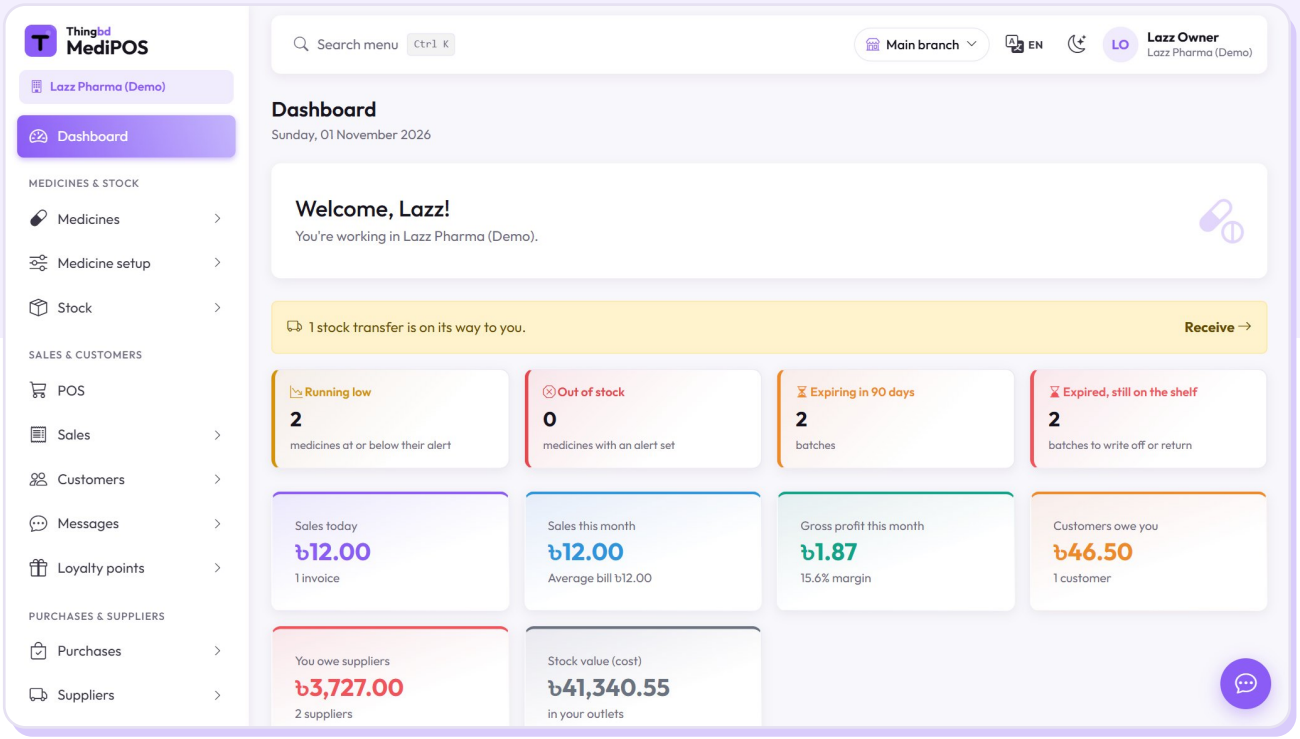
Thingbd

Software for pharmacies in Bangladesh

ThingBD MediPOS

The complete guide: what it does, how to install it on a shop computer or a server, and how to use it every day.



Sell

Barcode POS, batches and expiry, English or Bangla receipts

Run the shop

Stock, purchases, accounts, staff, messages and reports

Always ready

A ThingBD desktop icon, starts with Windows, sells offline

Version 1.0 · November 2026

thingbd.com

For pharmacy owners, their staff, and whoever installs MediPOS.

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Tip. Words in **bold** are buttons or menu items in MediPOS. A path such as **Shop setup** → **Invoice & printing** means: open **Shop setup** in the menu on the left, then click **Invoice & printing**. Press **Ctrl + K** anywhere to search the menu by name.

PART 1

Getting to know MediPOS

What it is, what it does, and what you need to run it.

1 About MediPOS

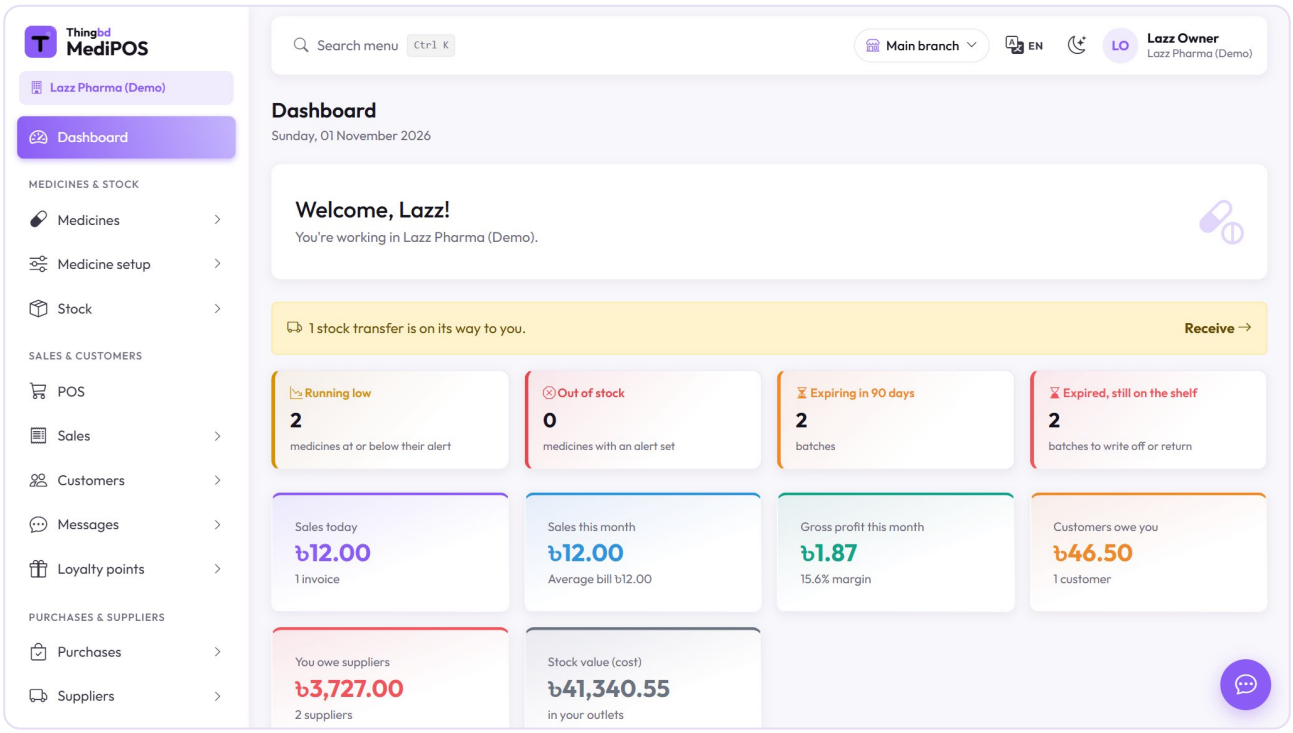
ThingBD MediPOS is a point-of-sale and back-office system made for medicine shops. One program covers the counter, the shelves and the office: sell and print receipts, keep stock by batch and expiry date, buy from suppliers, keep the accounts, run staff attendance and payroll, and message customers. Every screen works in **English and Bangla**, in a **light or dark** theme, on a computer, a tablet or a phone.

MediPOS can run in two ways. Pick the one that suits the shop:

Way to run it	Best for	Where the data lives
On the shop's own computer	One pharmacy with one or a few tills. Nothing to rent; keeps working when the internet is down. **Part 2** of this guide.	On that Windows PC (a database file in the MediPOS folder).
On an online server	ThingBD selling MediPOS to many pharmacies, or a chain whose branches sell from anywhere. **Part 4**.	On a Linux server with PostgreSQL, reached over HTTPS.

Three kinds of people use MediPOS:

- **Platform admin**: the owner of the MediPOS installation (ThingBD, or you on a shop PC). Adds pharmacies, sets plans and records subscription payments, sells SMS credits, keeps the shared medicine catalog and switches on the support assistant.
- **Pharmacy owner**: sees and changes everything in their pharmacy, in every branch.
- **Staff**: managers, cashiers and others. Each person has a role (Manager, Cashier, or roles you make) that decides which menus and buttons they see, and which branches they can work in.



The owner's dashboard: today's sales, profit, dues, stock alerts and a 30-day chart.

2 What MediPOS does

A short tour of every part of the program, in the order of the menu on the left.

Area	What you can do
Dashboard	Sales today and this month, average bill, gross profit, what customers owe you and what you owe suppliers, stock value, running-low and expiring alerts, transfers waiting, a 30-day sales chart, top sellers and recent sales. Staff check in and out here.
POS (till)	Full-screen counter: barcode scan or search by brand, generic or code; best sellers and dosage-form buttons; sell by piece, strip or box at retail or wholesale price; first-expiry-first-out batches; line and bill discounts, offers, VAT and rounding; split payment (cash, bKash, Nagad, bank, card); due sales with credit limits; loyalty points; hold and recall carts; quotations; receipts on 58 mm, 80 mm or A4.
POS without internet	When the connection drops the POS keeps selling from a copy kept in the browser, prints an offline receipt and sends the sales by itself when the connection is back.
Medicines	Medicine list with brand, strength, form, generic, maker, category, rack, units and pack sizes, prices, alert level and prescription flag. Add from the shared catalog, import from Excel, bulk price update, barcode labels.
Stock	Current stock per branch with value at cost and MRP, stock card per medicine, low stock with a suggested order, expiring and expired batches, opening stock import.
Inventory control	Transfers between branches, damage and expired write-offs, stock counts that correct the system to what is on the shelf.
Purchases & suppliers	Purchases with batch, expiry, free units and cost; purchase returns; supplier ledgers and payments.
Sales & customers	Sale list and detail, sale returns, cash registers with Z-reports, quotations, promotions, customer ledgers, due collection.

Area	What you can do
Messages	SMS, WhatsApp and email: the bill after a sale, due reminders, a daily stock alert and marketing campaigns. SMS through MediPOS SMS credits or the shop's own provider; an outbox shows every message.
Loyalty points	Customers earn points on what they spend and spend them at the POS as money off.
Accounts	Cash, bank and mobile-banking accounts; expenses and other income; deposits, withdrawals and transfers; double-entry books kept automatically: journal, profit & loss, balance sheet, cash flow, trial balance and account statements.
Audit	**Branch audits**: head office checks a branch for a period (sales, cash, stock, money and books), with a checklist, findings and a signed report. **Activity log**: who added, changed or deleted what, with the values before and after.
People	Employees with or without a login, roles and permissions, attendance (daily sheet, calendar, shifts, shop networks, self check-in), advances and monthly payroll with payslips.
Reports	About twenty reports (sales, profit, stock, purchases, customers, suppliers, staff, registers, messages, points) with filters, Excel download and print or PDF.
Shop setup	Business settings, invoice and printing design (with Bangla invoices and QR codes), branches and counters, data export and backup, subscription.
Platform	For the platform admin: pharmacies, plans and payments, SMS credits, the medicine catalog, the support assistant.
Everywhere	English and Bangla, light and dark theme, **Ctrl + K** menu search, works on phones, a help chat when it is switched on.

3 What you need

For a shop computer (Part 2)

- Windows 10 or 11, 64-bit. 4 GB of memory (8 GB is better) and 2 GB of free disk.
- Google Chrome or Microsoft Edge (Edge comes with Windows).
- Internet for the first install (it downloads about 100 MB), and later for SMS, email and updates. Selling does not need the internet.
- Python 3.13 (free). Section 4 shows how to install it.
- Optional: a 58 mm or 80 mm thermal receipt printer installed as a Windows printer, an A4 printer, and a USB barcode scanner (any scanner that types like a keyboard).

For an online server (Part 4)

- A Linux server (Ubuntu 24.04 LTS) with 2 CPU cores, 4 GB memory and 40 GB disk to start with.
- A domain name pointed at the server, for example **pos.yourcompany.com**.
- Someone comfortable with a Linux terminal for the first setup.

PART 2

Install on a Windows shop computer

Python, the installer, the ThingBD desktop icon, starting with Windows, and the installed app.

4 Install step by step

Step 1. Install Python

1. Open python.org/downloads/windows and download the latest **Python 3.13 Windows installer (64-bit)**.
2. Run it. On the first screen tick **Add python.exe to PATH**, then click **Install Now**.
3. When it says *Setup was successful*, click **Close**.

Step 2. Put the MediPOS folder on the computer

Copy the MediPOS folder you received from ThingBD (unzip it first if it came as a .zip) to `C:\MediPOS`. If you use Git, you can clone the repository there instead:

```
git clone <your MediPOS repository address> C:\MediPOS
```

Important. Leave the folder where you put it. The desktop icon and the automatic start point to this folder; if you move it, run the installer again (Step 3) to fix the icons.

Step 3. Run the installer

1. Open the folder `C:\MediPOS\scripts\windows`.
2. Double-click **Install-MediPOS.bat**. A black window opens and shows each step.
3. If Windows shows *Windows protected your PC*, click **More info**, then **Run anyway**. Do this only for the MediPOS installer you got from ThingBD.
4. Wait. The first run takes 2 to 5 minutes because it downloads the packages MediPOS needs.

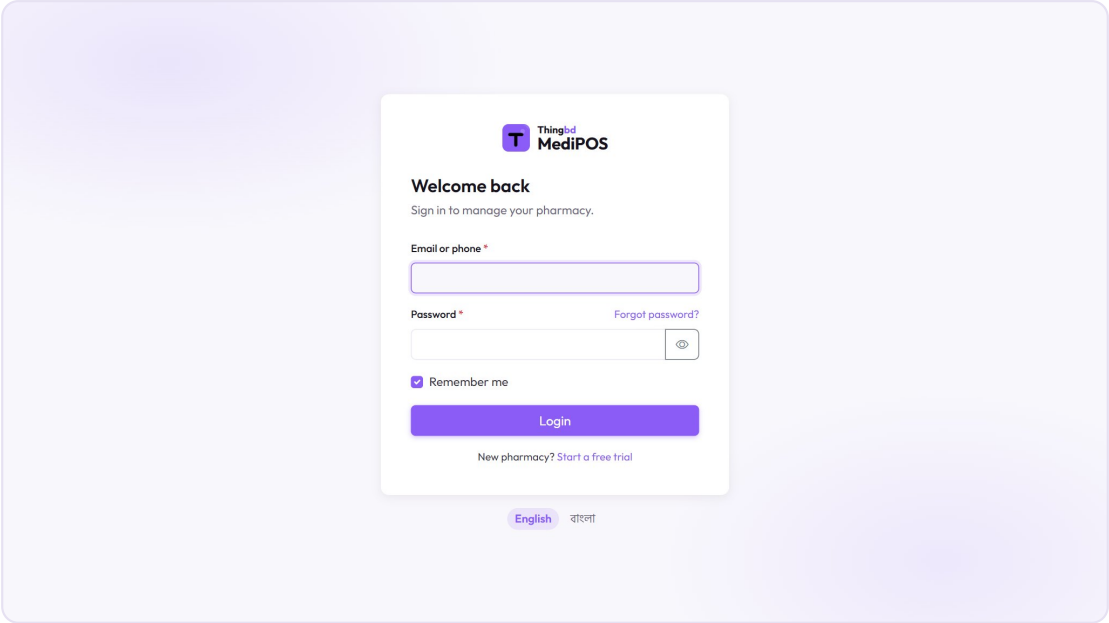
The installer does all of this by itself:

Step	What happens
Python	Finds Python 3.12 or newer and makes a private Python for MediPOS in `C:\MediPOS\venv`.
Packages	Installs everything listed in `requirements.txt` (Django, the Waitress web server, Excel and QR code libraries).
Settings	Writes a `.env` settings file with a new secret key, port 8800 and the shop-PC settings. An existing `.env` is kept as it is.
Database	Creates the database (`C:\MediPOS\db.sqlite3`) or brings it up to date after an update.
Platform admin	If there is no platform admin yet, asks for an email address and a password (twice). This is the master login: keep it safe.

Step	What happens
Shortcuts	Puts the ThingBD MediPOS icon on the desktop and in the Start menu, adds **Stop MediPOS** to the Start menu, and makes MediPOS start when you sign in to Windows.
Start	Starts MediPOS and opens it in its own window.

Step 4. Sign in

The MediPOS window opens on the sign-in page. Sign in with the platform admin email and password you just made, then follow **section 8, First-time setup**, to create your pharmacy and its owner login.



The sign-in page. The English / Bangla switch is under the box.

5 The desktop icon and starting automatically

Yes: MediPOS gets a proper icon **outside** its folder, with the ThingBD logo, and it starts by itself. The installer makes these shortcuts:

Shortcut	Where	What it does
ThingBD MediPOS	Desktop and Start menu	Starts MediPOS if it is not running, then opens the POS in its own window. If MediPOS is already running, it just opens the window.
ThingBD MediPOS	Windows Startup folder	Starts MediPOS quietly every time you sign in to Windows, without opening a window, so the till is ready and the installed app works straight away.
Stop MediPOS	Start menu → ThingBD MediPOS	Stops MediPOS on this computer (before an update, or to copy the database for a backup).

What happens when you double-click the icon

1. MediPOS starts in the background. There is no black window: the web server and the message sender run quietly.

2. After an update it brings the database up to date by itself.
3. It opens **http://localhost:8800/sales/pos/** in Chrome (or Edge) as an app window, without tabs or an address bar.
4. Every minute it sends waiting SMS and email, and once a day the stock alert and due reminders you switched on.

> **Good to know.** Closing the MediPOS window does **not** stop MediPOS; it keeps running in the background, which is what you want during the day. Use **Start → ThingBD MediPOS → Stop MediPOS** when you really want to stop it. Shutting down Windows stops it too.

Changing the shortcuts later

Run the installer script again from a Command Prompt opened in `C:\MediPOS`, with one of these switches:

```
powershell -ExecutionPolicy Bypass -File scripts\windows\install.ps1 -NoStartup
```

Switch	Effect
`-NoStartup`	Keeps the desktop and Start menu icons, but MediPOS no longer starts with Windows.
`-ShortcutsOnly`	Only makes the shortcuts again (for example after moving the folder or a new Windows user).
`-Remove`	Takes all the shortcuts away. The program and your data stay in the folder.

MediPOS answers on port **8800**. If another program already uses that port, MediPOS tells you so; set `MEDIPOS_PORT` (and `SITE_URL`) in the `.env` file to a free port, such as 8810, and start it again.

6 Install MediPOS as an app

MediPOS is an installable web app. Installed, it gets its own window and its own ThingBD icon in the Start menu and on the taskbar, and the POS opens even when the connection to MediPOS is lost.

In Google Chrome

1. Open **http://localhost:8800** and sign in.
2. In the address bar, click the **Install** icon (a small screen with a down arrow). Or open the three-dot menu and choose **Cast, save and share → Install page as app**.
3. Click **Install**. MediPOS opens in its own window and appears in the Start menu.

In Microsoft Edge

1. Open **http://localhost:8800** and sign in.
2. Open the three-dot menu, then **Apps → Install this site as an app**, and click **Install**.

To pin it, right-click the MediPOS icon on the taskbar while it is open and choose **Pin to taskbar**.

> **Tip.** The installed app is a window onto MediPOS; MediPOS itself still has to be running on the computer. With the automatic start from section 5 it always is, so you can open MediPOS from either icon.

7 Using MediPOS from other computers in the shop

One computer, the **main PC**, keeps the data. Other computers on the same shop network can use it through their browser:

1. On the main PC, open Command Prompt and type `ipconfig`. Note the **IPv4 Address**, for example 192.168.0.10.
2. Open `C:\MediPOS\.env` in Notepad. Change `MEDIPOS_HOST=127.0.0.1` to `MEDIPOS_HOST=0.0.0.0`, and add the address to `ALLOWED_HOSTS`, for example `ALLOWED_HOSTS=localhost,127.0.0.1,192.168.0.10`. Save.
3. Stop MediPOS (Start menu → Stop MediPOS) and start it again with the desktop icon. If Windows Firewall asks, allow access on **private networks**.
4. On the other computers open **http://192.168.0.10:8800** in Chrome or Edge and sign in.

Tip. Give the main PC a fixed IP address in your router so the address does not change. The main PC must be switched on while the others sell.

Important. Browsers only allow the offline POS and app install on **https** addresses or on the computer itself. Other computers reaching the main PC over plain http can sell only while the shop network is up. For several tills that must keep selling through any outage, use an online server with HTTPS (Part 4).

PART 3

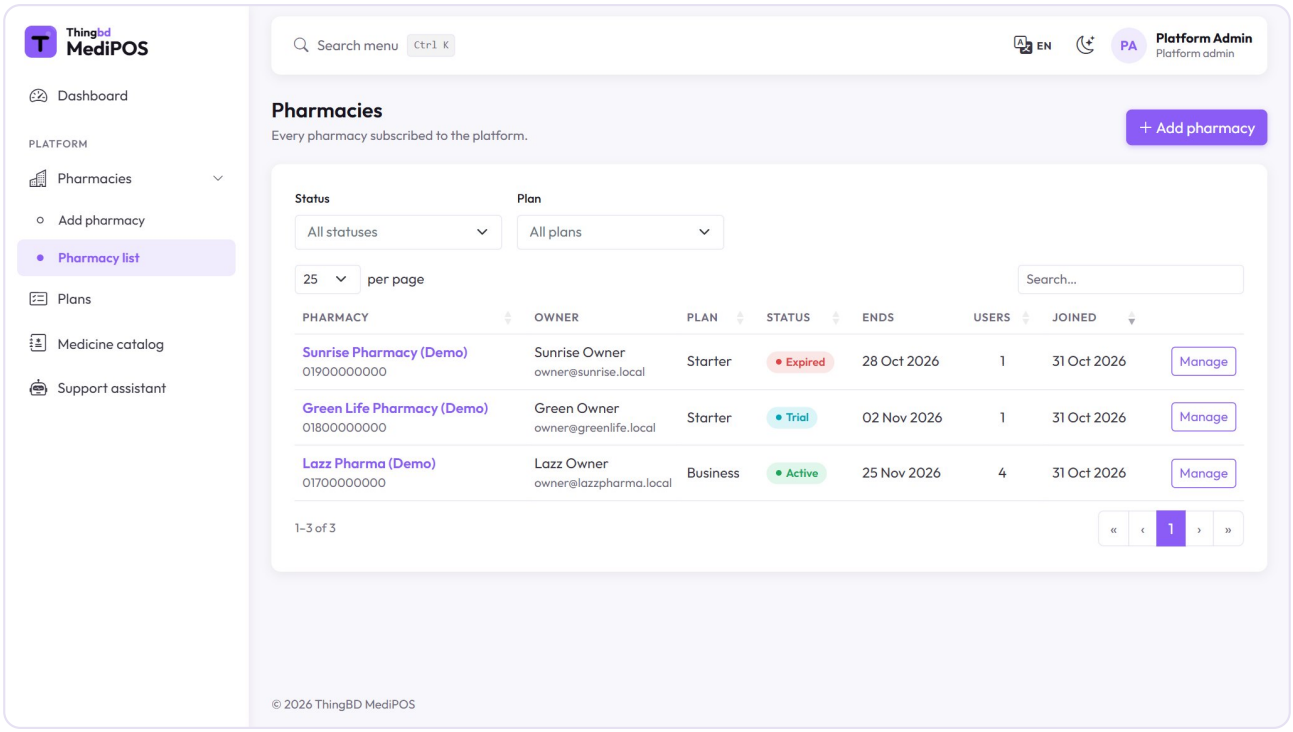
Using MediPOS

First-time setup, the POS, selling without internet, the daily routine, settings, backups and updates.

8 First-time setup

As the platform admin

1. Sign in with the platform admin login. Open **Pharmacies → Add pharmacy**.
2. Type the pharmacy's name and contact details, the owner's name, email or mobile number and a password, and choose a plan and the trial length. Save. MediPOS creates the pharmacy, its owner login and a **Main branch**. (A pharmacy can also sign itself up with **Start a free trial** on the sign-in page.)
3. Open the pharmacy from **Pharmacy list → Manage**. Here you **record payments** (each one extends the paid-until date), change the plan, extend a trial, add SMS credits, or suspend and reactivate the pharmacy.



Platform → Pharmacies: every pharmacy with its plan, status and end date.

Important. A pharmacy whose trial or paid period has ended (plus a few grace days) is locked out until a payment is recorded. On a shop PC you are the platform admin, so record a payment, or extend the trial, before it ends.

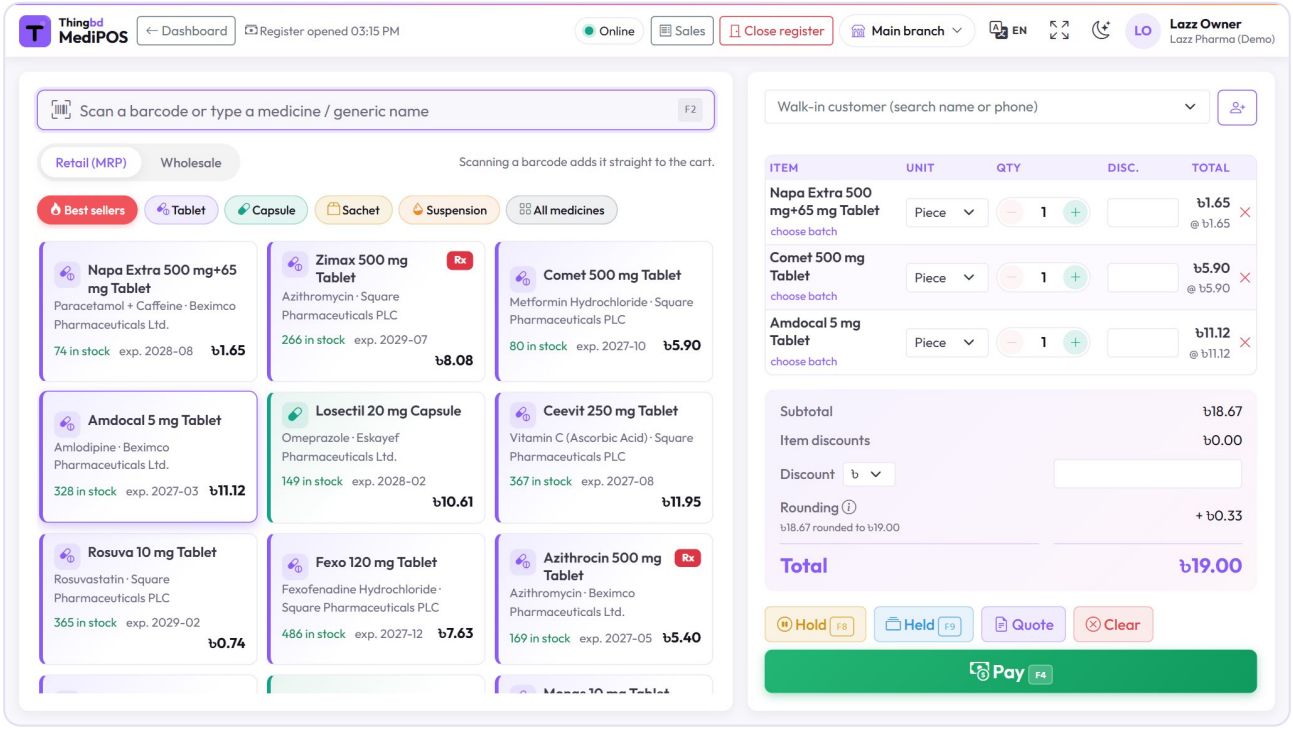
As the pharmacy owner

Sign out and sign in with the owner's login. The dashboard shows a **Set up your shop** checklist. Work through it in this order:

Step	Where	What to fill in
1. Your business	Shop setup → Business settings	Name, logo, drug license number, BIN/VAT number, currency and decimals, date format, invoice number prefix, default language.
2. Invoices	Shop setup → Invoice & printing	Paper (58 mm, 80 mm or A4), Bangla or English invoices, what to show, QR code, footer, and **Print the receipt after each sale**.
3. Branches	Shop setup → Outlets	The Main branch is already there. Add more branches and counters.
4. Money accounts	Accounts → Payment accounts	Cash, bKash and Nagad exist. Add bank accounts and set the opening balances.
5. Staff	People → Employees, Roles & permissions	Add each person with their role (Manager, Cashier, or your own) and the branches they work in.
6. Medicines	Medicines & stock → Medicines	**Add from catalog**, **Import from Excel**, or **Add medicine**. Set MRP, purchase price, units and the low-stock alert level.
7. Stock	Stock → Opening stock, or Purchases → Add purchase	Import what is on the shelves with batch and expiry, or receive your first purchase.
8. Customers	Customers → Customer list	Optional: regular customers with credit limits and default discounts. Walk-in customers need nothing.
9. Start selling	Sales & customers → POS	Open a cash register with the float in the drawer, and sell.

9 Selling at the POS

1. **Open a register.** The first time you open the POS it asks for the cash already in the drawer (the float).
2. **Find the medicine.** Scan the barcode, or type part of the brand, generic or code (press **F2** to jump to the search box). A scan adds the medicine straight to the cart. Before you type, the POS shows best sellers; the buttons switch to a dosage form or to all medicines.
3. **Adjust the cart.** Pick piece, strip or box, change the quantity, type a line discount. **Retail (MRP)** and **Wholesale** switch the price list. MediPOS takes the batch that expires first; **choose batch** overrides it. Offers that are running fill in the discount by themselves.
4. **Customer (optional).** Search by name or phone, or add a new one with the button next to the box. A sale on credit (due) needs a customer and stays within their credit limit. Customers with points can spend them here.
5. **Pay.** Press **Pay** or **F4**. Split the money across cash, bKash, Nagad, bank or card; MediPOS shows the change. Press **Complete**.
6. **Receipt.** It prints by itself if you switched that on; otherwise press **Print**. The bill can also go to the customer by SMS, WhatsApp or email.



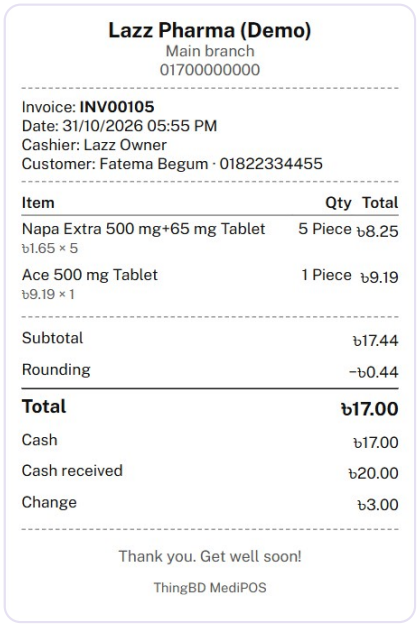
The POS with three medicines in the cart. Rounding to whole taka is switched on in this shop.

Other buttons and keys

- **Hold** (F8) parks the cart for the next customer; **Held** (F9) brings it back.
- **Quote** saves the cart as an A4 quotation without taking stock; later **Sell in POS** loads it again at today's prices.
- Medicines marked **Rx** ask the cashier to confirm the prescription was checked.
- **Close register** at the end of the shift: count the cash and MediPOS shows the Z-report with any shortage or excess.
- The **Online** badge on the top bar turns to **Offline** when the connection is lost (section 10).

Printer tips

- Choose the receipt width in **Shop setup** → **Invoice & printing**.
- In the print dialog, set **Margins: None** and **Scale: 100**. Chrome remembers it.
- In the printer's own settings in Windows, choose the paper width (58 or 80 mm).



An 80 mm receipt.

10 Selling without internet

On a **shop computer** (Part 2) MediPOS itself runs on the computer, so losing the internet does not stop anything: you keep selling as usual. Only SMS and email wait in the outbox and go out by themselves when the internet is back.

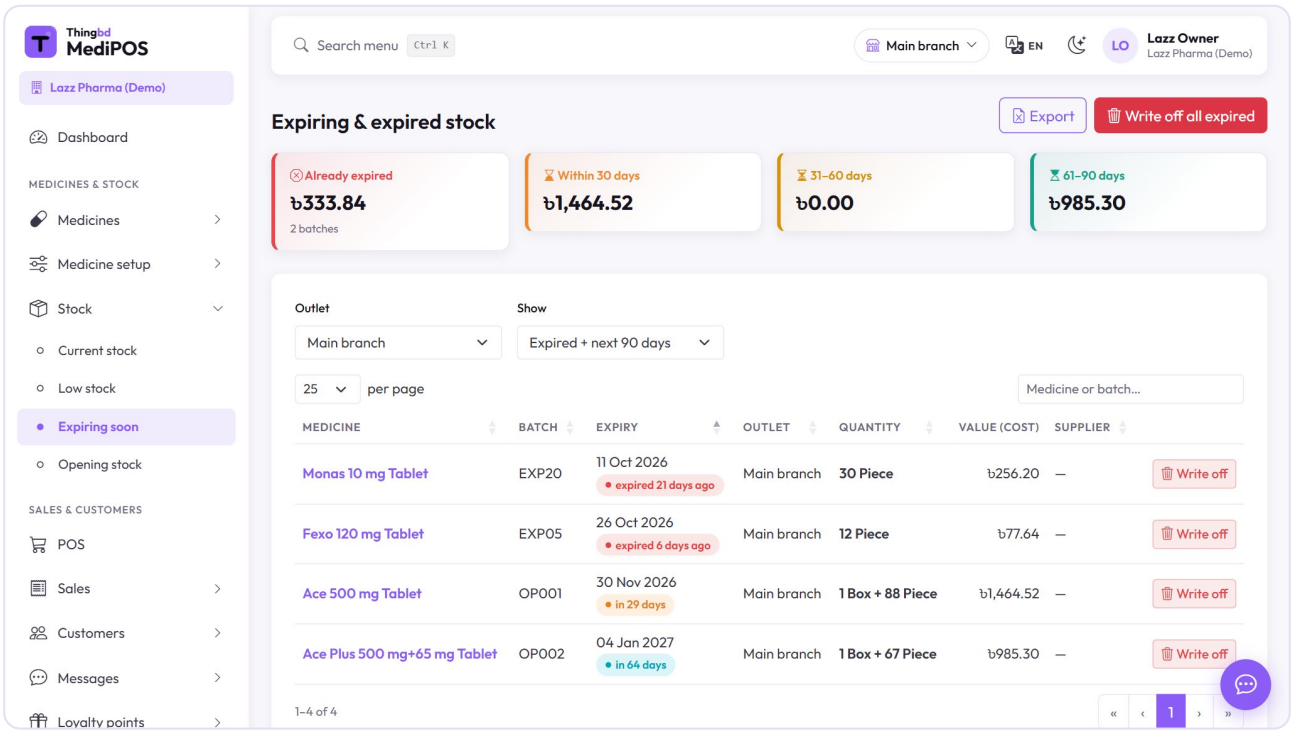
On an **online server** (Part 4), the POS keeps working when the connection drops:

- While online, the POS keeps a copy of its branch's medicines (prices, units, stock, barcodes, offers), the customers and the open register in the browser, refreshed every 10 minutes and after each sale.
- When the server cannot be reached, the badge turns to **Offline**. Search, scanning, best sellers and customers work from the copy. A completed sale gets a temporary number (**OFF-...**), the stock is taken off the copy, and an offline receipt prints.
- When the connection is back, the saved sales are sent by themselves (and every 30 seconds). Each is recorded with its real time and the note *Sold without internet*; a sale is never recorded twice.
- Click the badge to see sales waiting. If the server refuses one (stock gone, credit limit, a wrong computer clock) it shows why, with **Send again**, **Back to the cart** and **Delete**.
- Choosing a batch, holding carts, quotations and the other pages need the connection.

> **Tip.** Open the POS at least once while online on each till before you rely on it offline, so it has a fresh copy. Installing MediPOS as an app (section 6) makes the POS open even after the browser was closed during an outage.

11 The daily and monthly routine

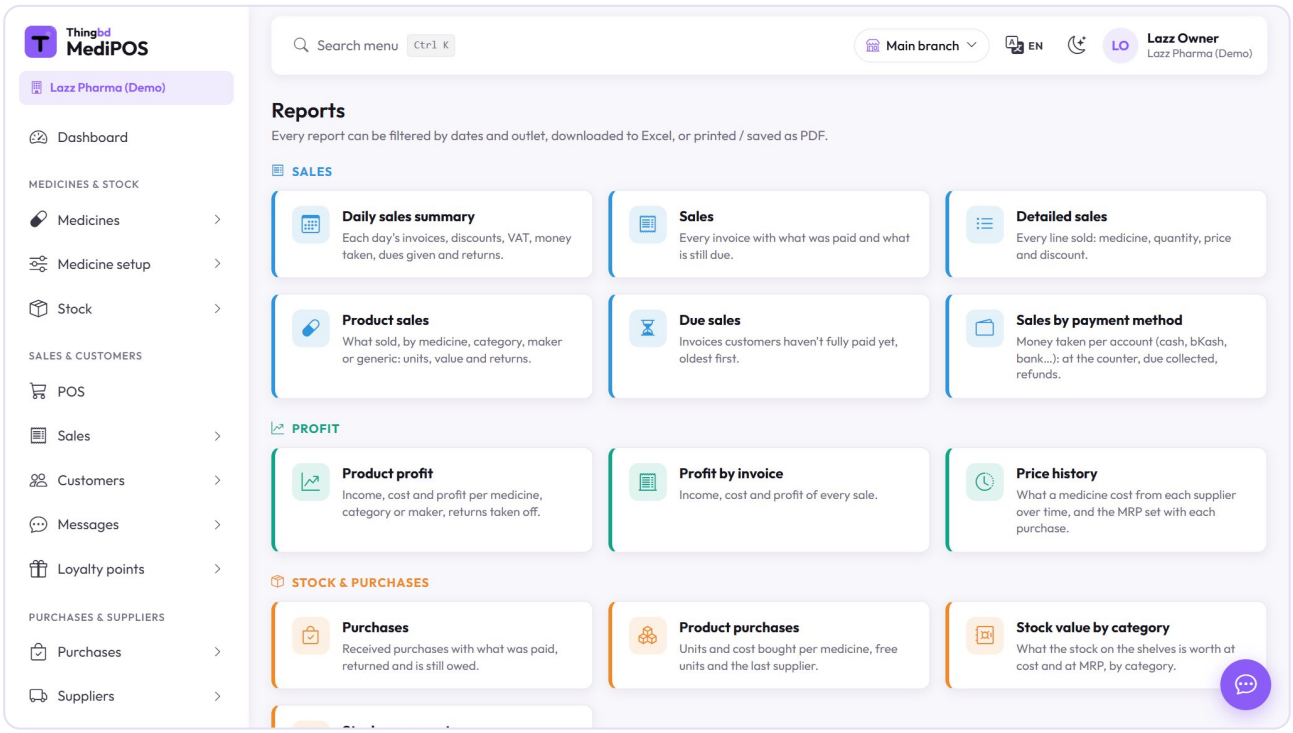
When	What to do	Where
Morning	Open the register with the float. Look at the dashboard alerts: running low, out of stock, expiring, transfers on the way.	POS, Dashboard
All day	Sell; collect dues; record expenses paid from the drawer; receive purchases.	POS, Customers → Collect due, Accounts → Expenses, Purchases
Evening	Close the register and check the Z-report. Make a backup (section 14).	POS → Close register, Shop setup → Data export & backup
Weekly	Order what is low; return or write off expiring stock; send due reminders.	Stock → Low stock / Expiring soon, Messages → Due reminders
Monthly	Count stock; run payroll; read profit & loss; audit branches; download a full backup.	Inventory control → Stock counts, People → Payroll, Accounts → Accounting, Accounts → Audit



Stock → Expiring soon: batches by how soon they expire, with Write off and Return to supplier.

12 Where to find things

To do this	Open
Sell	Sales & customers → POS
See or reprint a sale; take a return	Sales → Sale list; Sales → Sale returns
Cash registers and Z-reports	Sales → Cash registers
Quotations and promotions (offers)	Sales → Quotations; Sales → Promotions
Collect money a customer owes	Customers → Collect due
Add medicines	Medicines → Add medicine / Add from catalog / Import from Excel
Change many prices at once; print barcode labels	Medicines → Bulk price update; Medicines → Barcode labels
Categories, generics, makers, units, racks	Medicine setup
Stock now, low stock, expiring, opening stock	Stock → Current stock / Low stock / Expiring soon / Opening stock
Receive stock from a supplier	Purchases & suppliers → Purchases → Add purchase
Return to a supplier; pay a supplier	Purchases → Purchase returns; Suppliers → Supplier payments
Move stock between branches	Inventory control → Transfers
Write off damaged or expired stock	Inventory control → Damage & expired
Count the shelves	Inventory control → Stock counts
Expenses, other income, deposits	Accounts → Income & expenses
Profit & loss, balance sheet, cash flow, journal	Accounts → Accounting
Audit a branch; see who changed what	Accounts → Audit → Branch audits / Activity log
Staff, roles, attendance, payroll	People
SMS, WhatsApp, email and campaigns	Sales & customers → Messages
Loyalty points	Sales & customers → Loyalty points
All reports	Reports
Business details, invoice design, branches, backup, subscription	Shop setup

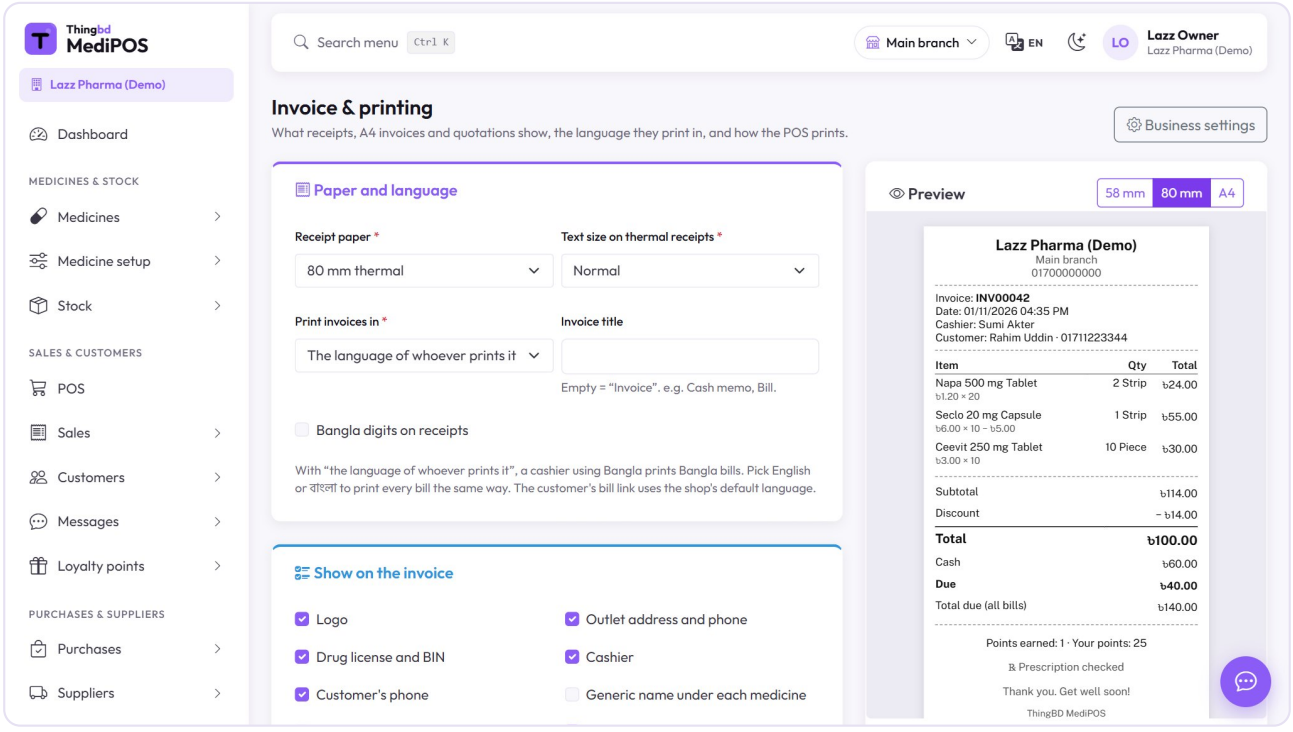


Reports: every report has filters, Excel download and print or PDF.

13 Settings you may want to change

Invoices and receipts

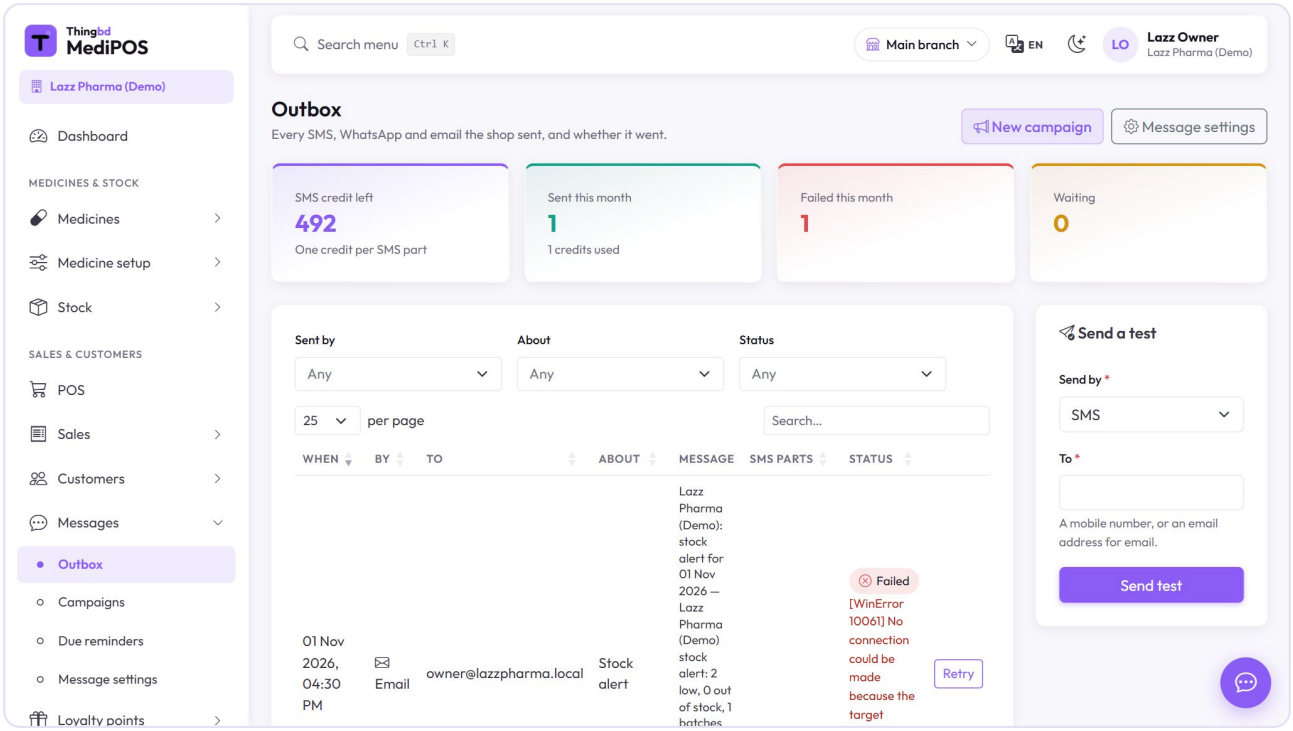
Shop setup → Invoice & printing decides what every receipt, A4 invoice and quotation shows, with a live sample on the right. Bills can print in the language of whoever prints them, always in English, or always in Bangla, with Bangla digits if you like. Add an A4 letterhead image, terms and conditions, and a QR code with the customer's bill link or your own text (website, bKash number).



Shop setup → Invoice & printing with the 80 mm sample.

Messages (SMS, WhatsApp, email)

- **Messages** → **Message settings**: choose how SMS go out (**MediPOS SMS credits** bought from the platform, your **own SMS provider**, or off), and switch on the bill after each sale, due reminders and the daily stock alert.
- Every message appears in **Messages** → **Outbox** as waiting, sent or failed (with the reason). **Retry** resends a failed one.
- WhatsApp buttons open the chat with the text already typed. Email goes through the mail server set in `.env` (`EMAIL_HOST` and the lines after it; see Appendix A).
- Messages need the internet. On a shop PC they wait and go out by themselves when it is back.



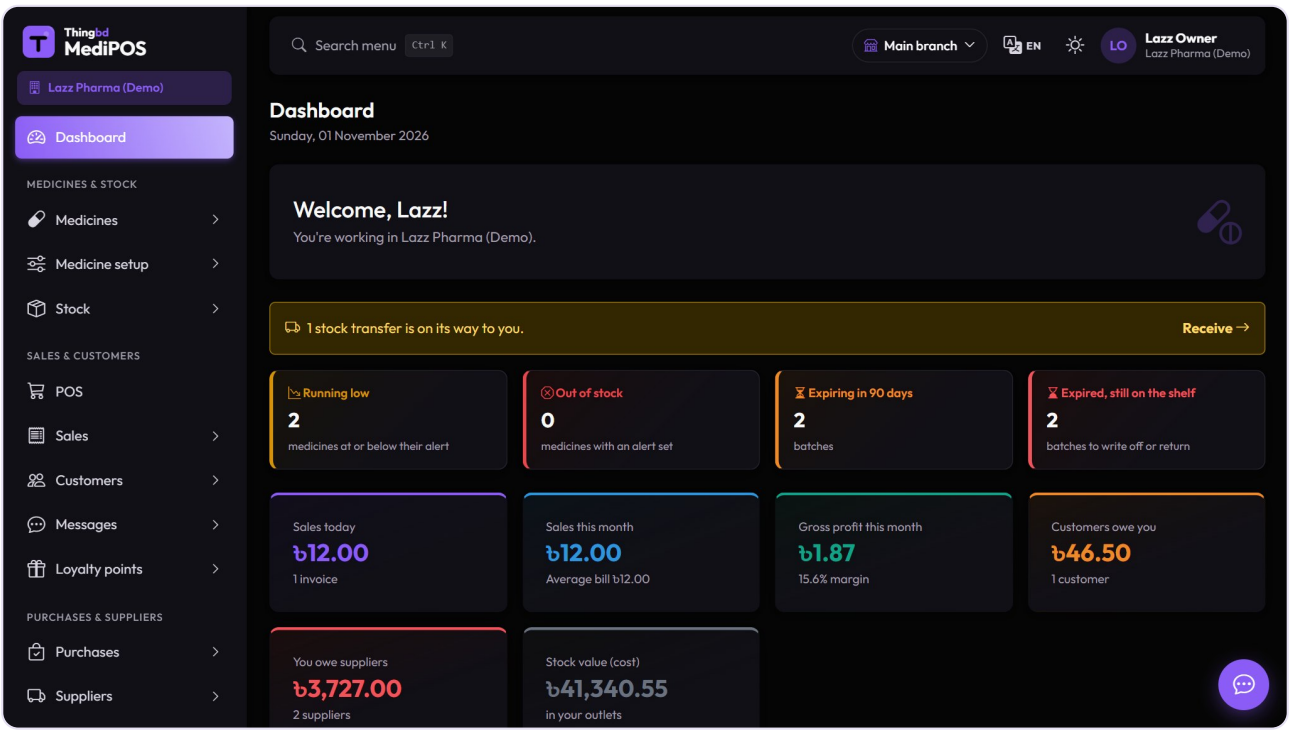
Messages → Outbox: every SMS, WhatsApp and email with its status.

Loyalty points

Loyalty points → **Loyalty settings**: how much a customer spends for one point, the smallest number of points they can spend, what a point is worth, and the most of a bill points can pay.

Language and theme

The **EN** button on the top bar switches between English and Bangla; the moon button switches between light and dark. Each person's choice is remembered. The shop's default language is in **Business settings**.



The dark theme: black with ThingBD purple, like thingbd.com.

Support assistant (help chat)

The platform admin switches it on in **Platform → Support assistant**. It starts with **built-in help**: free how-to answers in English and Bangla, no AI and nothing sent anywhere. To use an AI instead, choose any OpenAI-compatible service there (Groq's free plan, OpenRouter, Google Gemini, a local Ollama, or OpenAI), paste its key and set a daily limit per pharmacy. The chat button appears only for people whose role has the support permission (owners by default).

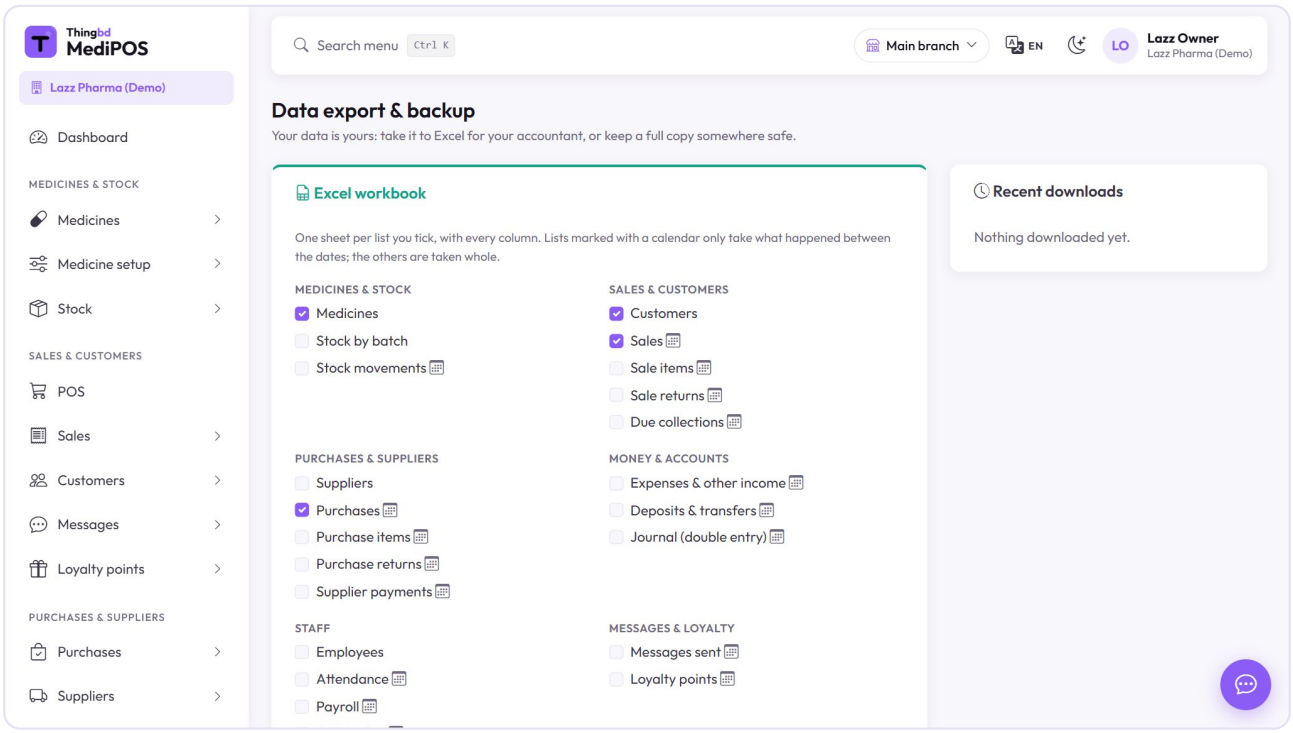
Roles and permissions

People → Roles & permissions lists every module with its actions (see, add, edit, delete and special ones such as profit reports or backups). Menus and buttons a person is not allowed to use disappear for them. Owners always have everything.

14 Backups and restoring

Shop setup → Data export & backup (owners, unless a role allows it) has two kinds of download. Every download is written in the activity log.

- **Excel workbook**: tick any of the 21 lists (medicines with stock, batches, sales, customers with their dues, purchases, suppliers, expenses, journal, staff, attendance, payroll and more) and a period.
- **Full backup (.zip)**: every record of the pharmacy plus its uploaded files. Restoring it is done by ThingBD support.



Shop setup → Data export & backup.

A complete copy of a shop computer

On a shop PC all data lives in the `C:\MediPOS` folder. To keep a copy you can put back yourself:

1. Stop MediPOS (Start menu → ThingBD MediPOS → Stop MediPOS).
2. Copy `db.sqlite3`, `.env` and the `media` folder from `C:\MediPOS` to a USB drive or a cloud folder, in a folder named with the date.
3. Start MediPOS again with the desktop icon.

To restore: stop MediPOS, copy those files back into `C:\MediPOS`, and start it. Do this at least every week, better every day, and keep several dates.

> **Important.** `.env` holds the secret key and any SMS or email passwords. Keep backups somewhere private.

15 Updating MediPOS

1. Make a backup (section 14).
2. Stop MediPOS (Start menu → Stop MediPOS).
3. Put the new version's files into `C:\MediPOS`, replacing the old ones, but **keep** `.env`, `db.sqlite3`, `media` and `logs`. If you installed with Git, run `git pull` in `C:\MediPOS` instead.
4. Double-click `scripts\windows\Install-MediPOS.bat` again. It updates the packages and the database, keeps your settings and data, and starts MediPOS.

PART 4

Install on an online server

For many pharmacies, or branches selling from different places: Ubuntu, PostgreSQL, Nginx and HTTPS.

16 Server install

This is a summary for the person who sets up the server. It uses Ubuntu 24.04, PostgreSQL for the database, Gunicorn to run MediPOS, Nginx in front of it and a free Let's Encrypt certificate for HTTPS. Replace **pos.example.com** with your domain and choose your own passwords.

1. Packages, database and code

```
sudo apt update
sudo apt install -y python3 python3-venv postgresql nginx certbot python3-certbot-nginx git
sudo -u postgres psql -c "CREATE USER medipos WITH PASSWORD 'a-strong-database-password';"
sudo -u postgres psql -c "CREATE DATABASE medipos OWNER medipos;"
sudo adduser --system --group --home /srv/medipos medipos
sudo chmod 755 /srv/medipos
sudo -u medipos git clone <your MediPOS repository address> /srv/medipos/app
cd /srv/medipos/app
sudo -u medipos python3 -m venv venv
sudo -u medipos venv/bin/pip install -r requirements.txt gunicorn
```

2. Settings: /srv/medipos/app/.env

```
DJANGO_SETTINGS_MODULE=config.settings.prod
SECRET_KEY=<a long random text, made with the command below>
DEBUG=False
ALLOWED_HOSTS=pos.example.com
CSRF_TRUSTED_ORIGINS=https://pos.example.com
SITE_URL=https://pos.example.com
DATABASE_URL=postgres://medipos:a-strong-database-password@localhost:5432/medipos
BEHIND_PROXY=True
MESSAGES_SEND_NOW=False
DEFAULT_FROM_EMAIL=MediPOS <no-reply@example.com>
EMAIL_HOST=smtp.example.com
EMAIL_PORT=587
EMAIL_HOST_USER=no-reply@example.com
EMAIL_HOST_PASSWORD=<mail password>
EMAIL_USE_TLS=True
SMS_GATEWAY_URL=<your SMS provider's URL with {to} {message} {sender} {api_key}>
```

Make the secret key with `python3 -c "import secrets; print(secrets.token_urlsafe(50))"` and keep the file private: `sudo chown medipos /srv/medipos/app/.env` and `sudo chmod 600 /srv/medipos/app/.env`.

3. Database tables, files and the first admin

```
cd /srv/medipos/app
export DJANGO_SETTINGS_MODULE=config.settings.prod
sudo -u medipos -E venv/bin/python manage.py migrate
sudo -u medipos -E venv/bin/python manage.py collectstatic --noinput
sudo -u medipos -E venv/bin/python manage.py createsuperuser
```

Important. `manage.py` uses the development settings unless `DJANGO_SETTINGS_MODULE` is set, so always export it (as above) before running commands on the server.

4. Keep MediPOS and the message sender running (systemd)

Create **/etc/systemd/system/medipos.service**:

```
[Unit]
Description=ThingBD MediPOS
After=network.target postgresql.service

[Service]
User=medipos
WorkingDirectory=/srv/medipos/app
Environment=DJANGO_SETTINGS_MODULE=config.settings.prod
ExecStart=/srv/medipos/app/venv/bin/gunicorn config.wsgi:application \
    --bind 127.0.0.1:8000 --workers 3 --timeout 60
Restart=always

[Install]
WantedBy=multi-user.target
```

And **/etc/systemd/system/medipos-messages.service**, which sends SMS and email every minute:

```
[Unit]
Description=ThingBD MediPOS messages
After=medipos.service

[Service]
User=medipos
WorkingDirectory=/srv/medipos/app
Environment=DJANGO_SETTINGS_MODULE=config.settings.prod
ExecStart=/srv/medipos/app/venv/bin/python manage.py send_messages --loop 60
Restart=always

[Install]
WantedBy=multi-user.target
```

```
sudo systemctl daemon-reload
sudo systemctl enable --now medipos medipos-messages
```

5. Nginx and HTTPS

Create **/etc/nginx/sites-available/medipos**, link it into **sites-enabled**, then ask Let's Encrypt for the certificate:

```
server {
    server_name pos.example.com;
    client_max_body_size 20M;

    location /media/ { alias /srv/medipos/app/media/; }
    location / {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
    }
}
```

```
sudo ln -s /etc/nginx/sites-available/medipos /etc/nginx/sites-enabled/
sudo nginx -t && sudo systemctl reload nginx
sudo certbot --nginx -d pos.example.com
```

6. Backups and updates

- Back up every night with `pg_dump` (for example from cron: `pg_dump -U medipos medipos | gzip > /backups/medipos-$(date +%F).sql.gz`) and copy `/srv/medipos/app/media` and `.env`. Keep copies on another machine.
- To update: `git pull`, `pip install -r requirements.txt`, `migrate`, `collectstatic --noinput` (with `DJANGO_SETTINGS_MODULE` exported), then `sudo systemctl restart medipos medipos-messages`.
- Docker packaging, scheduled backups and monitoring are planned for a later phase.

PART 5

Help and reference

Fixing common problems, every setting in .env, the folder layout and useful commands.

17 Troubleshooting

Problem	What to do
The desktop icon does nothing	The first start can take 10 to 20 seconds. If nothing opens, look at `logs\medipos.log` and `logs\desktop.log` in the MediPOS folder, or run `venv\Scripts\python.exe scripts\windows\medipos_desktop.py` in a Command Prompt in that folder to see the error.
"Another program is using port 8800"	Set `MEDIPOS_PORT` and `SITE_URL` in `.env` to a free port (such as 8810) and start MediPOS again. Install the app again from the new address.
The installer says Python was not found	Install Python 3.13 with **Add python.exe to PATH** ticked (section 4), close the window and run the installer again.
"Running scripts is disabled on this system"	Start the installer with **Install-MediPOS.bat**, not the .ps1 file; the .bat allows the script for that one run.
Installing packages fails	Check the internet connection (and any proxy), then run the installer again; it continues where it stopped.
Forgot a password	Staff: the owner resets it in **People** → **Employees**. Anyone with email set up: **Forgot password?** on the sign-in page. Otherwise, in a Command Prompt in the MediPOS folder: `venv\Scripts\python.exe manage.py changepassword their@email.com`.
"Your subscription has ended"	The platform admin opens the pharmacy and records a payment or extends the trial (section 8).
Receipts print too small, too big or with margins	Check the paper size in **Invoice & printing**, set **Margins: None** and **Scale: 100** in the print dialog, and the paper width in the printer's settings.
Offline sales are waiting and not sent	Click the **Offline** badge. If it asks you to sign in, sign in again; if a sale was refused, the reason is shown with Send again, Back to the cart and Delete.
SMS or email not arriving	**Messages** → **Outbox** shows failed messages and why: no credits, a wrong gateway address, mail settings in `.env`, or no internet.
MediPOS is slow with many tills on one PC	One PC with SQLite suits a shop with a few tills. For more, or for branches, move to an online server with PostgreSQL (Part 4); ThingBD can move the data.

Appendix A: settings in .env

The `.env` file in the MediPOS folder holds settings that are not on the screens. Open it with Notepad, change a line, save, then stop and start MediPOS.

Setting	What it does	Shop PC
`DJANGO_SETTINGS_MODULE`	Which settings to use: `config.settings.desktop` (shop PC), `config.settings.prod` (server).	desktop
`SECRET_KEY`	Signs sessions and bill links. Long and random; never share it.	made for you
`MEDIPOS_PORT`	The port MediPOS answers on.	8800
`MEDIPOS_HOST`	127.0.0.1 = this computer only; 0.0.0.0 = other computers on the network too.	127.0.0.1
`ALLOWED_HOSTS`	Addresses MediPOS answers to, separated by commas.	localhost, 127.0.0.1
`CSRF_TRUSTED_ORIGINS`	Server only: the https address, such as https://pos.example.com.	not needed
`SITE_URL`	The start of links in SMS and email (the customer's bill).	http://localhost:8800
`DATABASE_URL`	Empty = SQLite file `db.sqlite3`. A postgres:// address on a server.	empty
`TRIAL_DAYS`, `SUBSCRIPTION_GRACE_DAYS`	Length of a new pharmacy's trial, and the days allowed after a paid period ends.	14, 3
`SUBSCRIPTION_PAYMENT_INSTRUCTIONS`	The "how to pay" text pharmacies see.	optional
`MESSAGES_SEND_NOW`	True = send SMS/email while the page saves; False = leave them to the worker (every minute).	False
`DEFAULT_FROM_EMAIL`, `EMAIL_HOST`, `EMAIL_PORT`, `EMAIL_HOST_USER`, `EMAIL_HOST_PASSWORD`, `EMAIL_USE_TLS`	The mail server for emailed bills and password resets. Gmail: smtp.gmail.com, 587, TLS and an app password.	optional
`SMS_GATEWAY_URL`, `SMS_GATEWAY_METHOD`, `SMS_GATEWAY_API_KEY`, `SMS_GATEWAY_SENDER`, `SMS_GATEWAY_SUCCESS_TEXT`	The platform's SMS provider for MediPOS SMS credits. Empty = SMS are only logged.	optional
`BEHIND_PROXY`	Server only: True when Nginx passes the visitor's address (used for attendance check-in).	False
`SUPPORT_AI_API_KEY`	The AI key for the support assistant, if not typed in the platform settings.	optional

Appendix B: the MediPOS folder

Folder or file	What is in it
`apps\`	The program, one folder per part (sales, stock, accounting, messaging…).
`config\settings\`	Settings: `desktop.py` (shop PC), `prod.py` (server), `dev.py` and `test.py`.
`scripts\windows\`	`Install-MediPOS.bat`, `install.ps1` and `medipos_desktop.py` (the launcher the icon runs).
`static\`, `templates\`, `locale\`	Screens, styles, the ThingBD icons, and the Bangla translation.
`docs\`	This guide and the build plan.
`venv\`	MediPOS's own Python, made by the installer.
`db.sqlite3`	**Your data** on a shop PC.
`media\`	**Your uploaded files**: logo, letterhead, medicine photos.
`.env`	**Your settings** and secret key.
`logs\`	`medipos.log` (errors and starts), `desktop.log`, `medipos.pid`.

Appendix C: useful commands

Open a Command Prompt in `C:\MediPOS` (in the folder, type `cmd` in the address bar and press Enter). Set the shop-PC settings first:

```
set DJANGO_SETTINGS_MODULE=config.settings.desktop

rem Start MediPOS and open the POS, or stop it
venv\Scripts\pythonw.exe scripts\windows\medipos_desktop.py
venv\Scripts\pythonw.exe scripts\windows\medipos_desktop.py --stop

rem Set a new password, or add another platform admin
venv\Scripts\python.exe manage.py changepassword someone@example.com
venv\Scripts\python.exe manage.py createsuperuser

rem Send waiting SMS and email now, or re-post the accounts from the documents
venv\Scripts\python.exe manage.py send_messages
venv\Scripts\python.exe manage.py rebuild_books
```

Questions or problems: contact ThingBD at thingbd.com. MediPOS also answers *How do I…?* questions in its help chat when the platform admin has switched it on.